

# Spectroscopy as a Probe of Spacetime Foam

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Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources. We show that this jitter modifies the first-order temporal coherence  $g^{(1)}(\tau)$  and hence the observed spectrum, providing a spectroscopic probe complementary to existing imaging-channel (PSF, Strehl) constraints. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by  $e^{-\sigma_\phi^2}$  and an incoherent pedestal, yielding a spectral shape-distortion bound on the Ng–Perlman foam exponent  $\alpha$ . Applied to the MeV emission lines of GRB 221009A in the dynamical-foam limit ( $\tau_{\text{foam}} \ll \tau_c$ ) we obtain  $\alpha \gtrsim 0.63$ . For the broadband CMB the foam induces an  $e^{-\sigma_\phi^2(\nu)}$  deficit ( $\sigma_\phi^2 \propto \nu^2$ ) with a low-frequency pedestal; a Fisher analysis of COBE/FIRAS reaches  $\alpha \lesssim 0.52$  under the physical Planck-scale pedestal kernel (essentially zero in-band;  $\Delta\alpha \approx 0.01$  loosening under an adversarial band-filling template, modulo a  $\Delta\alpha \sim 0.03$  cosmological-redshift systematic); the cosmological-accumulation variant tightens the FIRAS threshold to  $\alpha \lesssim 0.55$ , with PIXIE projected to reach  $\alpha \lesssim 0.54$  from instrumental noise improvement alone. The final FIRAS threshold awaits a first-principles derivation of the foam temporal autocorrelation  $C_\alpha(\tau)$ , which determines the pedestal shape and hence the foam- $\mu$  degeneracy that currently dominates the error budget. None of these bounds exclude the holographic value  $\alpha = 2/3$ , which would require a confirmed narrow spectral line at  $E_{\text{obs}} \gtrsim 720$  MeV in a  $z \sim 0.15$  GRB.

## I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length  $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$  m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the  $\alpha$ -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where  $D$  is the proper distance to the source,  $\alpha$  parametrizes the foam model, and  $A_\alpha$  is a dimensionless coefficient of order unity. A photon traversing distance  $D$  passes through approximately  $N \sim D/\ell_P$  independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent  $\alpha$  controls how the per-cell displacements accumulate: for random-walk foam ( $\alpha = 1/2$ ), contributions add in quadrature ( $\sigma_\ell \propto \sqrt{D\ell_P}$ ); holographic foam ( $\alpha = 2/3$ ) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ( $\alpha = 1$ ) yields a distance-independent  $\sigma_\ell \propto \ell_P$ , corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic spacetime foam to date has been the imaging channel. Spacetime foam accumulates a cumulative phase dispersion  $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^\alpha/\lambda$  across the entrance aperture, degrading PSF sharpness and

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erasing interference fringes once  $\Delta\phi \gtrsim 2\pi$  [7]. For PKS 1413+135 ( $\ell = 1.216$  Gpc) observed with HST at  $\lambda = 1.6 \mu\text{m}$ , the random-walk model ( $\alpha = 1/2$ ) predicts  $\Delta\phi \sim 10 \times 2\pi$ , so the detection of Airy rings in this image marginally rules out that model; the holographic model ( $\alpha = 2/3$ ) predicts  $\Delta\phi \sim 10^{-9} \times 2\pi$  and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

The most luminous GRB ever recorded, GRB 221009A at  $z = 0.151$ , has been examined in the imaging channel by Steinbring [11] and Steinbring & Ng [12]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [11] showed that the holographic ( $\alpha = 2/3$ ) model predicts an angular widening below  $\sim 1^\circ$ , consistent with the Fermi and Swift localization data for GRBs at  $z \leq 5$  up to peak energies of  $\sim 10$  MeV. Steinbring & Ng [12] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with  $\alpha = 2/3$ . These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A.

We develop a framework based on the first-order temporal coherence function  $g^{(1)}(\tau)$  of light from astrophysical sources. This function is complementary to the first-order spatial coherence function  $g^{(1)}(B)$ , which forms the basis of imaging-channel probes. The first-order coherence — the field–field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance  $\sigma_\phi^2$  grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation  $C_\alpha(\tau)$ , and the distribution of differential jitter. Section III derives the first-order coherence function  $g_{\text{obs}}^{(1)}(\tau)$  of the observed field and establishes the Gaussian closed form. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem, establishing the split of the observed spectrum into a coherent line core suppressed by  $e^{-\sigma_\phi^2}$  and a broad pedestal, and connects them to the imaging Strehl ratio. Section V applies the spectral shape-distortion bound to two sources: GRB 221009A (Section V A) and the CMB as observed by COBE/FIRAS (Section V B). Section VI summarizes the main results and outlines future observational directions.

## II. SPACETIME FOAM MODEL

### A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where  $E_{\text{src}}(t)$  is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for the thermal and incoherent non-thermal sources considered here, including GRB high-energy emission and the CMB) and  $\delta t(t)$  is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of  $E_{\text{src}}$ . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance  $D$ :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (3)$$

The vacuum  $c$  in Eq. (3) reflects the standard Ng–Perlman treatment of foam as a small perturbation atop a flat (or FLRW) background, with photons travelling at  $c$  along stochastically perturbed geodesics. In the geometric-optics limit the path-length shift  $\Delta \ell_{\text{tot}}$  is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as  $\lambda^2$ . This achromaticity distinguishes foam from other astrophysical screens — such as plasma dispersion, interstellar scintillation, and atmospheric turbulence — all of which imprint frequency-dependent signatures. For broadband sources such as the CMB, the foam distortion is likewise distinct from standard spectral distortions ( $\mu, y$ ) that have different frequency dependences.

### B. Temporal Decorrelation of Spacetime Foam

The scalar variance  $\sigma_\ell^2(D)$  (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — determines how the arrival-time jitter  $\delta t(t)$  along a single line of sight is correlated across the lag  $\tau$ . Quantum foam models posit that each Planck-scale cell along the line of sight fluctuates on a characteristic timescale  $\tau_{\text{foam}}$ , so that the jitter at times  $t$  and  $t + \tau$  is correlated for  $\tau \ll \tau_{\text{foam}}$  and essentially independent for  $\tau \gg \tau_{\text{foam}}$ . We introduce  $C_\alpha(\tau; D)$  as the normalized *temporal* autocorrelation of the integrated path-length fluctuation  $\Delta \ell_{\text{tot}}(t)$ , i.e.  $C_\alpha(\tau; D) \equiv \langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle / \sigma_\ell^2(D)$ , which inherits the temporal structure of the per-cell foam dynamics over the  $\sim D/\ell_P$  cells crossed by the line of sight. Because  $C_\alpha$  is defined on the *integrated* path, its temporal shape inherits a residual dependence on  $D$  from the way per-cell foam dynamics combine along the line of sight; the explicit  $D$ -argument records this dependence. Equivalently,

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (4)$$

The timescale  $\tau_{\text{foam}}$  is an additional free parameter of the foam model, not fixed by the  $\alpha$ -model variance relation alone. In the standard phenomenological framework [6, 13], each of the  $\sim D/\ell_P$  foam cells along the line of sight independently adopts a new configuration on timescale  $\tau_P \equiv \ell_P/c \approx 5 \times 10^{-44}$  s; the integrated path-length jitter  $\delta \ell_{\text{tot}}(t)$  — a sum of  $\sim D/\ell_P$  independent contributions — likewise decorrelates on timescale  $\tau_{\text{foam}} \sim \tau_P$ . With  $C_\alpha(\tau) \approx 0$  for all nonzero accessible lags, the foam is fully decorrelated on any observable timescale — the most optimistic scenario for detection — and for any astrophysical source considered here ( $\tau_c/\tau_{\text{foam}} \gtrsim 10^{22}$  even at 37 MeV) the analysis sits deep within the dynamical regime. Models with long-range *temporal* correlations — the temporal analogue of the long-range *transverse* spatial correlations of Ref. [10] that evade interferometric imaging bounds — would give  $C_\alpha(\tau) \approx 1$  over macroscopic timescales and thereby drive  $\sigma_\Delta^2(\tau) \rightarrow 0$ , suppressing *both* the GRB spectroscopic signal of Section V A and the CMB spectral distortion of Section V B. The two channels therefore probe the same temporal decorrelation regime, and both depend on the dynamical-foam assumption  $\tau_{\text{foam}} \ll \tau_c$  adopted here.

### C. Differential Jitter and Its Distribution

With the temporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the differential jitter between photons arriving at times  $t$  and  $t + \tau$ :

$$\Delta_t(\tau) \equiv \delta t(t + \tau) - \delta t(t). \quad (5)$$

For fixed  $D$ , stationarity implies that the distribution of  $\Delta_t(\tau)$  depends only on  $\tau$ ; we suppress the  $t$  subscript and write  $\Delta(\tau)$  henceforth. Expanding the square and applying Eq. (4):

$$\begin{aligned} \sigma_\Delta^2(\tau) &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau) - \Delta \ell_{\text{tot}}(t)]^2 \rangle \\ &= \frac{1}{c^2} [\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2 C_\alpha(\tau)] \\ &= \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)]. \end{aligned} \quad (6)$$

By the central limit theorem (CLT),  $\Delta(\tau)$  is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}(0, \sigma_\Delta^2(\tau)), \quad (7)$$

where  $\sigma_\Delta^2$  is taken from Eq. (6); for any source at  $D \gtrsim 1$  kpc, the cell count  $N_{\text{cells}} \sim D/\ell_P \gtrsim 10^{54}$  amply satisfies the large- $N$  requirement and higher-cumulant corrections are negligible. For  $\alpha = 1/2$  (random walk), per-cell path-length variances are equal ( $\ell_P^2$  each) and the CLT applies in its standard form. For  $\alpha \neq 1/2$ , the per-cell contributions scale differently with  $D$  and the equality of per-cell variances is a model assumption rather than a direct CLT consequence; Gaussianity is adopted as the minimal assumption consistent with the  $\alpha$ -model.

The characteristic function of  $\Delta(\tau)$  at carrier frequency  $\omega_0$  defines a foam suppression factor applicable to any source profile. Since  $\Delta(\tau)$  is Gaussian,

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2} \equiv \mathcal{F}_{\text{foam}}(\tau), \quad (8)$$

with  $\mathcal{F}_{\text{foam}}(0) = 1$ .

From the foam path-length variance  $\sigma_\ell^2(D_C)$  and carrier frequency  $\omega_0$ , we define  $\sigma_\phi^2 \equiv \frac{1}{2}\omega_0^2 \cdot (2\sigma_\ell^2/c^2) = (\omega_0\sigma_\ell/c)^2$  as the asymptotic phase variance — the value  $\frac{1}{2}\omega_0^2\sigma_\Delta^2(\tau)$  approaches as  $C_\alpha(\tau; D) \rightarrow 0$ . In the dynamical-foam limit, the Ng–Perlman path-length model (1) with  $D_C$  the line-of-sight comoving distance [4] gives

$$\sigma_\phi^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}}\right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}. \quad (9)$$

Substituting  $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$  from Eq. (6), Eq. (8) takes the compact form

$$\mathcal{F}_{\text{foam}}(\tau) = e^{-\sigma_\phi^2[1 - C_\alpha(\tau; D)]}. \quad (10)$$

At  $\tau \rightarrow \infty$ :  $\mathcal{F}_{\text{foam}}(\infty)$  equals  $e^{-\sigma_\phi^2}$  in the dynamical limit ( $C_\alpha \rightarrow 0$ ) and approaches unity for quasi-static foam ( $C_\alpha \approx 1$ ). Equation (9) uses observer-frame wavelength  $\lambda_{\text{obs}}$  throughout and so neglects the cosmological redshift of the photon along the propagation path: at redshift  $z'$  the photon's local wavelength was  $\lambda(z') = \lambda_{\text{obs}}/(1+z')$ , and the phase accumulated per unit path length scales as  $1/\lambda^2$ . For a model in which foam path-length variance accumulates uniformly in comoving distance, the redshift-weighted form is

$$\sigma_\phi^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}}\right)^2 \int_0^{z_s} (1+z')^2 \frac{d\sigma_\ell^2}{dD_C} \frac{dD_C}{dz'} dz', \quad (11)$$

which reduces to Eq. (9) when  $(1+z')^2$  can be taken constant over the path. For the GRB at  $z = 0.151$  the path-averaged  $\langle(1+z')^2\rangle_{D_C} \approx 1.16$ , enhancing  $\sigma_\phi^2$  by 16% — negligible at the precision of the present analysis. For the CMB at  $z_{\text{LS}} \approx 1089$  the weighting is dominated by the high- $z$  portion of the path: with Planck 2018 flat  $\Lambda$ CDM the size and sign of this correction depend on how foam fluctuations evolve in cosmic time; two natural anchorings bracket the range: (a) variance accumulating uniformly in comoving distance,  $d\sigma_\ell^2/dD_C = \text{const}$ , enhances  $\sigma_\phi^2$  by a factor of  $\sim 10^4$ ; (b) variance anchored to proper-now coordinates with foam frozen at the present epoch gives  $(1+z')^2 \rightarrow 1$  inside the integral, leaving  $\sigma_\phi^2$  unchanged. Bounds quoted below use Eq. (9) with  $A_\alpha = 1$  and adopt anchoring (b); the CMB systematic of Section VB discusses the corresponding shift under anchoring (a).

### III. FIRST-ORDER COHERENCE

#### A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t+\tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (12)$$

Substituting Eq. (2) and letting  $u = t - \delta t(t)$ ,  $v = t + \tau - \delta t(t + \tau)$ , so that  $v - u = \tau - \Delta(\tau)$ , stationarity of  $E_{\text{src}}$  gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (13)$$

Since  $\delta t$  is independent of  $E_{\text{src}}$ , and since  $\langle |E_{\text{obs}}(t)|^2 \rangle = \langle |E_{\text{src}}(t - \delta t)|^2 \rangle = \langle |E_{\text{src}}|^2 \rangle$  by stationarity (so the normalisation in Eq. (12) cancels), averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (14)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

#### B. Gaussian Source Profile

The GRB 221009A emission line is well described by a Gaussian profile in each epoch [14], and it serves as the representative case admitting the simplest closed form. A Gaussian line of  $1\sigma$  width  $\sigma_\nu$  centred on  $\nu_0$  corresponds via the Wiener–Khinchin theorem to the coherence function

$$g_{\text{src}}^{(1)}(\tau) = e^{-i\omega_0\tau} e^{-\tau^2/2\tau_G^2}, \quad (15)$$

where  $\omega_0 \equiv 2\pi\nu_0$  and  $\tau_G \equiv 1/(2\pi\sigma_\nu)$  is the Gaussian coherence decay time. The carrier factor  $e^{-i\omega_0\tau}$  encodes the line's position in the spectrum and must be retained: dropping it would substitute a baseband Gaussian centred at  $\nu = 0$  for the physical line and would lose the dominant foam effect (line-peak suppression). Substituting into the jitter-convolution integral (14) and completing the square gives the exact closed form

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma_\Delta^2(\tau)}} \exp\left[-i\omega_0\tau \frac{\tau_G^2}{\tau_G^2 + \sigma_\Delta^2(\tau)}\right] \times \exp\left[-\frac{\tau^2 + \omega_0^2 \tau_G^2 \sigma_\Delta^2(\tau)}{2[\tau_G^2 + \sigma_\Delta^2(\tau)]}\right]. \quad (16)$$

The observed coherence is again a Gaussian-modulated oscillation, but with three foam-induced modifications: (i) a slight redshift of the carrier,  $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / [\tau_G^2 + \sigma_\Delta^2(\tau)]$ ; (ii) a wider Gaussian decay of  $g_{\text{obs}}^{(1)}(\tau)$  in lag (decay scale  $\sqrt{\tau_G^2 + \sigma_\Delta^2(\tau)}$  vs.  $\tau_G$ ); and (iii) a global amplitude suppression by the factor  $\exp[-\omega_0^2 \tau_G^2 \sigma_\Delta^2 / \{2(\tau_G^2 + \sigma_\Delta^2)\}]$ . Normalisation  $g_{\text{obs}}^{(1)}(0) = 1$  holds because  $\sigma_\Delta(0) = 0$ . The narrowband limit  $\tau_G \rightarrow \infty$  of Eq. (16) reduces to a pure phase-modulated oscillation, which is the single-mode building block from which the broadband CMB spectrum of Section V B 2 is constructed by incoherent superposition.

#### IV. IMPLICATIONS FOR SPECTROSCOPY

##### A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (14) with closed forms Eq. (16) — carries direct spectroscopic content: for a stationary field the power spectral density  $S(\nu)$  and  $g^{(1)}(\tau)$  are Fourier-transform pairs, implying

$$S_{\text{obs}}(\nu) = \mathcal{F}\left\{g_{\text{obs}}^{(1)}(\tau)\right\}(\nu). \quad (17)$$

The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and because the foam-induced jitter is a stationary process independent of the source (foam acts on photons after emission, while  $\delta t(t)$  statistics are time-translation invariant on observational timescales). The observed field is then stationary when the integration time exceeds the foam coherence time  $\tau_{\text{foam}}$  (the lag scale beyond which  $C_\alpha(\tau; D) \rightarrow 0$ ), so that the time-averaged spectrum coincides with its ensemble average:  $T_{\text{int}} \gg \tau_{\text{foam}}$ .

In the dynamical limit  $C_\alpha(\tau; D) \rightarrow 0$  for  $|\tau| \gtrsim \tau_{\text{foam}}$ ,  $\mathcal{F}_{\text{foam}}(\tau)$  acquires a long-lag plateau  $e^{-\sigma_\phi^2}$  outside a narrow window  $|\tau| \lesssim \tau_{\text{foam}}$ . Fourier-transforming via Eq. (17), the plateau produces a coherent core  $e^{-\sigma_\phi^2} S_{\text{src}}(\nu)$ , with the complementary fraction  $1 - e^{-\sigma_\phi^2}$  of the source flux in a pedestal sourced by the short-lag window. The per-mode derivation in Section V B 2 makes this decomposition explicit for a broadband source.

For the Gaussian source profile of Section III B, Fourier-transforming Eq. (16) with  $\sigma_\Delta^2$  held at its asymptotic value  $2\sigma_\ell^2/c^2$  gives the explicit Gaussian core

$$S_{\text{obs}}^{\text{core}}(\nu) = \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma_\Delta^2}} \exp\left[-\frac{\sigma_\phi^2 \tau_G^2}{\tau_G^2 + \sigma_\Delta^2}\right] \cdot \frac{1}{\sqrt{2\pi} \sigma_{\text{obs}}} \exp\left[-\frac{(\nu - \nu_0^{\text{eff}})^2}{2\sigma_{\text{obs}}^2}\right], \quad (18)$$

with  $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / (\tau_G^2 + \sigma_\Delta^2)$  and  $\sigma_{\text{obs}} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}] < \sigma_\nu$ . The Gaussian's carrier shift, line narrowing, and peak suppression are the frequency-domain images of items (i)–(iii) of Section III B.

The pedestal in either decomposition is sourced by the short-lag window  $|\tau| \lesssim \tau_{\text{foam}}$  and so spreads over a frequency range  $\sim 1/\tau_{\text{foam}}$ . When  $\tau_{\text{foam}} \ll \tau_G, \sigma_\Delta$  the pedestal is correspondingly broad and shallow, and the coherent core is the operative spectrum near the line.

##### B. Connection to the Imaging Strehl Ratio

The first-order coherence function  $g_{\text{obs}}^{(1)}(\tau)$  derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation  $C_\alpha(\tau; D)$  of Eq. (10). The Strehl ratio of an imaging telescope is the *spatial* counterpart: phase variations across the aperture plane are governed by an aperture-plane

coherence function  $C_\alpha^{\text{sp}}(B; D)$ , constructed by direct analogy to the temporal  $C_\alpha(\tau; D)$  of Section II B under the same foam-cell statistics.

The structural equivalence is not merely formal: in both cases the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable. Temporally, the variable is the differential jitter  $\Delta(\tau) = \delta t(t + \tau) - \delta t(t)$  with variance  $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$  (Eq. (6)). Spatially, the variable is the differential phase  $\Delta\phi(B) = \phi(\mathbf{x} + \mathbf{B}) - \phi(\mathbf{x})$  between two points on the aperture plane separated by baseline  $B$ , with variance

$$\sigma_{\Delta\phi}^2(B) = 2\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]. \quad (19)$$

The focal-plane field of a telescope is the Fraunhofer Fourier transform of the aperture-plane field, giving the diffraction-limited PSF when the aperture phase is uniform. Foam imprints a Gaussian random phase  $\phi(\mathbf{x})$  on the aperture with variance  $\sigma_\phi^2$  and two-point correlation  $C_\alpha^{\text{sp}}(B; D)$ , blurring this PSF. In the resolved limit  $C_\alpha^{\text{sp}}(D_{\text{tel}}; D) \rightarrow 0$  — where aperture phases at distinct points decorrelate — the on-axis peak is attenuated by the Strehl ratio

$$S \simeq e^{-\sigma_\phi^2} \quad (20)$$

(extended-Maréchal [4]), the surrounding Airy fringes are washed out by the same factor, and the complementary  $1 - e^{-\sigma_\phi^2}$  of the source flux scatters into an incoherent halo.

Imaging and spectroscopic channels each probe  $\sigma_\phi^2$  through a Fourier transform — aperture position  $\leftrightarrow$  focal-plane angle in imaging (Fraunhofer), lag  $\leftrightarrow$  frequency in spectroscopy (Wiener-Khinchin) — with the rapid chromatic tightening  $\sigma_\phi \propto 1/\lambda$  exploited in both Ng & Perlman’s [4] Table 1 and the analyses of Section V A. The shared  $\sigma_\phi^2$  that suppresses the coherent diffraction-limited PSF in imaging also suppresses the coherent line core in spectroscopy, placing the published imaging constraints [4, 7–9] on a common footing with the present framework. The duality is not perfect at the observational interface: isolating the foam contribution requires an independently characterized telescope aperture in the imaging channel and an independently characterized intrinsic source spectrum in the spectroscopic channel.

## V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling  $\sigma_\phi \propto 1/\lambda$ . Two observable consequences follow. First, individual spectral features are distorted according to the Gaussian-source result Eq. (18), with a carrier shift, line narrowing, and peak suppression that scale chromatically as  $\sigma_\phi \propto 1/\lambda$ . Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton- $y$  distortions. In this section we consider these two possibilities.

### A. GRB 221009A

For spectroscopy of a single source without an absolute flux reference, the smooth suppression factor  $e^{-\sigma_\phi^2(\nu)}$  attenuates line and continuum equally at any given frequency, preserving the equivalent width: uniform dimming is undetectable from the spectrum alone. The carrier-included closed form Eq. (16) supplies the needed shape-distortion criterion: foam broadens the observed line width by a calculable energy-dependent amount (derived below as Eq. (21)), and multi-epoch line-width measurements across a wide energy range give additional leverage, as we now show for GRB 221009A.

GRB 221009A (“the BOAT,” Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy  $E_{\text{iso}} \sim 10^{55}$  erg at redshift  $z = 0.151$ , a once-in- $\sim 10^4$ -year event [15]. Two independent spectral analyses have reported a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron-positron pair annihilation [14, 16], at observed energies spanning 6–37 MeV. This pair-annihilation identification is one proposed interpretation and is not yet definitive; the foam analysis depends only on Doppler-invariance of the line’s fractional width, not on the specific emission mechanism. We exploit the *multi-epoch* constancy of the fractional line width across this factor-of-six energy range to obtain a stringent constraint. Adopting flat  $\Lambda$ CDM ( $\Omega_M = 0.27$ ,  $\Omega_\Lambda = 0.73$ ,  $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ), the comoving distance is  $D_C(z = 0.151) \approx 0.63 \text{ Gpc} \approx 1.93 \times 10^{25} \text{ m}$  and  $\ln(D_C/\ell_P) \approx 138.3$ .

The analysis below operates in the dynamical-foam regime  $\tau_{\text{foam}} \ll \tau_c$  adopted throughout Section II; in the opposite quasi-static limit the framework yields no foam imprint on the line spectrum and the bound does not apply (caveat (vi) below).

*Multi-epoch line data.* Zhang et al. [14] perform a joint analysis of GECAM-C and Fermi/GBM data covering the full prompt emission, including the initial bright phase that saturated GBM alone. They detect a Gaussian emission line with centroid evolving as  $E_{\text{line}} \propto t^{-1}$  from  $\approx 37$  MeV down to  $\approx 6$  MeV over  $T_{\text{ref}} + 246\text{--}360$  s. In the seven sub-intervals where the line is sufficiently bright to constrain its width, the measured fractional width  $r_{\text{obs}} \equiv \sigma_{\text{obs}}/E_{\text{line}} = 0.10 \pm 0.01$  is consistent with a single constant across the full dynamic range ( $\chi^2/\text{dof} = 4/7$ ; Zhang et al. §4.4 and their Figure 7c). This constancy is the predicted behaviour of a Doppler-boosted line. In the comoving frame of the emitting plasma the line has fractional width  $r' = \sigma'/E'$ , fixed by the comoving emission physics (pair-plasma temperature, velocity dispersion). When the outflow has bulk Doppler factor  $\mathcal{D}$  — epoch-dependent as the jet decelerates — both the line centroid and its width are multiplied by the same factor:  $E = \mathcal{D}E'$ ,  $\sigma_{\text{int}} = \mathcal{D}\sigma'$ , so  $r_{\text{int}} \equiv \sigma_{\text{int}}/E = r'$ . The Doppler factor cancels in the ratio;  $r_{\text{int}}$  is a Lorentz-invariant property of the line, independent of epoch. The observed  $E_{\text{line}} \propto t^{-1}$  centroid evolution over a factor of  $\approx 6$  in energy traces the bulk-Doppler deceleration of the jet, and the simultaneous constancy of  $r_{\text{obs}}$  confirms the prediction. The line is independently detected by Ravasio et al. [16] in Fermi/GBM data alone, with  $\sigma_{\text{gauss}} = 1.31^{+0.31}_{-0.30}$  MeV at  $E_{\text{line}} = 12.56^{+0.30}_{-0.31}$  MeV in  $T + 280\text{--}300$  s ( $\sigma/E \approx 10.4\%$ , consistent with Zhang's  $r_{\text{obs}} = 0.10 \pm 0.01$  at  $0.4\sigma$  in the time interval that overlaps Zhang's bright-line window). Ravasio also reports the line in three later intervals ( $T + 300\text{--}360$  s) where  $\sigma/E$  rises to  $\approx 15\text{--}20\%$ , so the constancy of  $r_{\text{obs}}$  across the full 9–37 MeV lever arm rests on the GECAM-C+Fermi/GBM joint analysis [14], with Ravasio providing an instrument-independent confirmation of the line at the highest-signal-to-noise epoch.

*Width decomposition.* Because  $r_{\text{int}}$  is the same in every epoch, the multi-epoch measurements constrain a single common  $r_{\text{int}}$  alongside the foam amplitude  $r_{\text{foam},0}$ , giving the lever arm its statistical power. The Gaussian-source spectrum Eq. (18) gives the observed centroid and absolute width  $\nu_0^{\text{eff}} = \nu_0 \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$  and  $\sigma_{\text{obs}} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}]$ . Although  $\sigma_{\text{obs}}$  is slightly *narrower* than the intrinsic  $\sigma_\nu$  by a factor  $\tau_G/\sqrt{\tau_G^2 + \sigma_\Delta^2}$ , the centroid is redshifted by the *square* of that factor, so the fractional width that Zhang et al. report as  $\sigma/E$  in each epoch *increases*:

$$r_{\text{obs}}^2(E) = r_{\text{int}}^2 [1 + \sigma_\Delta^2/\tau_G^2(E)] = r_{\text{int}}^2 + r_{\text{foam}}^2(E), \quad (21)$$

with foam excess  $r_{\text{foam}}^2(E) \equiv r_{\text{int}}^2 \sigma_\Delta^2/\tau_G^2(E)$ . The relation is multiplicative in  $(1 + \sigma_\Delta^2/\tau_G^2)$ , not additive in independent foam/intrinsic widths: the foam acts on the source through the jitter-convolution Eq. (14), a *multiplication* of the spectrum by a Gaussian factor rather than a convolution with it (Section III B).

In the dynamical-foam limit ( $\tau_{\text{foam}} \ll \tau_c$ ),  $\sigma_\Delta^2(\tau)$  saturates to  $2\sigma_\ell^2/c^2$  on lags  $|\tau| \gtrsim \tau_{\text{foam}}$ . The broadening term in Eq. (21) is set by this variance at the source coherence lag  $\tau \sim \tau_G$ , which satisfies  $\tau_G \sim \tau_c \gg \tau_{\text{foam}}$  in this regime, so the saturated value is the operative one. The dominant foam effect on the spectrum is then carrier-phase suppression of the line peak by  $e^{-\sigma_\phi^2(E)}$ , together with line-width broadening captured by item (ii) of Section III B; both follow from Eq. (16). Substituting  $\sigma_\Delta = \sqrt{2}\sigma_\ell/c$  and  $\tau_G(E) = \lambda_{\text{obs}}(E)/(2\pi c r_{\text{int}})$ , the foam-to-intrinsic timescale ratio takes the compact form

$$\frac{\sigma_\Delta}{\tau_G(E)} = \sqrt{2} r_{\text{int}} \sigma_\phi(E), \quad (22)$$

the product of the intrinsic fractional width  $r_{\text{int}}$  (a source property) and the asymptotic phase variance  $\sigma_\phi$  (Eq. (9)). Substituting into Eq. (21) gives  $r_{\text{obs}}^2(E) = r_{\text{int}}^2 [1 + 2 r_{\text{int}}^2 \sigma_\phi^2(E)]$ , so the foam excess is

$$r_{\text{foam}}^{\text{(dyn)}}(E) = \sqrt{2} r_{\text{int}}^2 \sigma_\phi(E) = r_{\text{int}}^2 \cdot \frac{2\sqrt{2}\pi\sigma_\ell E}{hc} \propto E. \quad (23)$$

The  $r_{\text{int}}^2$  prefactor reflects two geometric factors: one converts the absolute foam-induced shift ( $\propto \sigma_\phi \nu_0$ ) into a fractional width; the other enters through  $\tau_G \propto 1/r_{\text{int}}$  in the broadening term  $\sigma_\Delta^2/\tau_G^2$ . A narrower intrinsic line (smaller  $r_{\text{int}}$ , longer  $\tau_G$ ) is therefore more robust to dynamical foam:  $\sigma_\Delta$  falls further below the source coherence time.

The fit-ready two-parameter model is

$$r_{\text{obs}}^2(E) = r_{\text{int}}^2 + r_{\text{foam},0}^2 (E/E_0)^2, \quad (24)$$

with  $r_{\text{foam},0} \equiv r_{\text{foam}}(E_0)$  the foam contribution at the reference energy  $E_0 = 37$  MeV ( $\lambda_0 = 3.4 \times 10^{-14}$  m).

*Fit and upper limit on  $r_{\text{foam},0}$ .* A  $\chi^2$  fit of the seven Zhang et al. measurements ( $E_i, r_i \pm \delta r_i$ ) to Eq. (24) with free parameters ( $r_{\text{int}}, r_{\text{foam},0} \geq 0$ ) yields a best fit at  $r_{\text{foam},0} = 0$ ,  $r_{\text{int}} = 0.111$ , with  $\chi_{\text{min}}^2 = 5.3$  ( $\chi^2/\text{dof} = 0.88$  with 6 effective degrees of freedom; since  $r_{\text{foam},0} = 0$  at the boundary the fit reduces to one free parameter  $r_{\text{int}}$ ), confirming the constant- $r_{\text{obs}}$  hypothesis adopted by Zhang et al. (their §4.4 and Figure 7c). Profiling over  $r_{\text{int}}$  gives the one-sided

95% CL upper limit  $r_{\text{foam},0}^{\text{max}} = 0.27$  at  $\Delta\chi^2 = 2.71$ . [17] The constraint is comparatively weak because the foam signal peaks at  $E_0 = 37$  MeV, where the single most uncertain data point (G1) resides. With seven measurements and per-point fractional uncertainties of 30–50%, the  $\Delta\chi^2 = 2.71$  coverage relies on the Gaussian- $\chi^2$  approximation; a bootstrap or profile-likelihood Monte Carlo would refine the exact coverage but is not expected to shift the bound by more than a few percent.

*Constraint on  $\alpha$ .* Eq. (23) gives  $\sigma_\ell = r_{\text{foam},0} \lambda_0 / (2\sqrt{2} \pi r_{\text{int}}^2)$ ; converting via  $\sigma_\ell^2 = D_C^{2(1-\alpha)} \ell_P^{2\alpha}$  (Eq. (1) with  $A_\alpha = 1$ ) gives

$$\alpha \gtrsim \frac{\ln(D_C/\lambda_0) + |\ln r_{\text{foam},0}^{\text{max}}| - |\ln(2\sqrt{2} \pi r_{\text{int}}^2)|}{\ln(D_C/\ell_P)} = \frac{89.2 + 1.3 - 3.3}{138.3} \approx 0.63, \quad (25)$$

where  $|\ln(0.27)| \approx 1.3$  and  $|\ln(2\sqrt{2} \pi r_{\text{int}}^2)| \approx 3.3$  for  $r_{\text{int}} = 0.066$ , the extreme value within the 95% CL contour. The cosmological redshift correction of Section III B enhances  $\sigma_\phi^2$  by 16% at  $z = 0.151$ , shifting the bound by  $\Delta\alpha \sim 5 \times 10^{-4}$  — negligible at the current precision. Using  $\sigma_\Delta = \sqrt{2} \sigma_\ell / c$  and  $c\tau_G(E_0) = \hbar c / (r_{\text{int}} E_0) \approx 8.1 \times 10^{-14}$  m for  $r_{\text{int}} = 0.066$ , the ratio  $\sigma_\Delta / \tau_G(E_0) \approx 4.1$  at the bound ( $\sigma_\ell \approx 2.3 \times 10^{-13}$  m), placing the bound deep in the strong-foam regime  $\sigma_\Delta \gtrsim \tau_G$ . The individual line-width measurements carry 30–50% fractional uncertainties; the multi-epoch lever arm is essential for reaching this precision.

The G1 interval ( $T+246$ – $256$  s,  $E \approx 37$  MeV) lies at or beyond the upper boundary of the GBM BGO detector range (4–35 MeV), so its line-width measurement carries larger systematic uncertainty than the later intervals. Refitting the remaining six intervals gives a best fit at  $r_{\text{foam},0} = 0$ ,  $r_{\text{int}} = 0.109$ , with  $\chi_{\text{min}}^2 = 4.4$  over five effective degrees of freedom, and a 95% CL profile boundary at  $r_{\text{foam},0}^{\text{max}} = 0.27$  paired with  $r_{\text{int}} = 0.066$  — yielding  $\alpha \gtrsim 0.63$ , statistically indistinguishable from the all-seven-interval bound. G1’s large fractional uncertainty ( $\sim 40\%$ ) gives it little weight in the  $\chi^2$  despite its position at the top of the lever arm.

The bound corresponds to  $\sigma_\Delta \gtrsim \tau_G$ , where Eq. (16) predicts *two* observable effects: (a) line-width broadening  $r_{\text{obs}}^2 = r_{\text{int}}^2 (1 + \sigma_\Delta^2 / \tau_G^2)$  — which the fit constrains via  $r_{\text{foam},0}^{\text{max}} = 0.27$  — and (b) a line-center redshift  $\nu_0^{\text{eff}} / \nu_0 = \tau_G^2 / (\tau_G^2 + \sigma_\Delta^2)$  that becomes substantial at the upper limit ( $\nu_0^{\text{eff}} / \nu_0 \approx 0.06$  at  $\sigma_\Delta / \tau_G \approx 4.1$ ). The lever-arm fit uses observed line centers as if they were intrinsic (pair-annihilation Doppler-boosted to  $E_i$  in each epoch) and does not separately constrain (b). Closing the gap requires either (i) an independent constraint on the per-epoch bulk Doppler factor  $\mathcal{D}_i$  — from afterglow blastwave modeling, TeV-photon photosphere arguments, or variability-timescale estimates — that pins the pre-foam centroid, or (ii) detection of a second emission line of known rest-frame energy whose distinct  $\tau_G(E)$  would produce a different foam-redshift factor in the same epoch and thereby separate  $\mathcal{D}_i$  from the foam shift. Neither is available in the present dataset. The bound is therefore weaker than what a joint fit would yield, and at the upper limit it is moreover not internally self-consistent: with  $\sigma_\Delta^2 / \tau_G^2 \propto E^2$ , the foam-induced centroid factor  $\nu_0^{\text{eff}} / \nu_0$  would vary across the data range from  $\approx 0.06$  at  $E = 37$  MeV to  $\approx 0.7$  at  $E = 6$  MeV, producing a strongly energy-dependent depression of the observed centroid that is visibly inconsistent with the smooth  $E_{\text{line}} \propto t^{-1}$  Doppler-deceleration trend in the data. A joint center-plus-width fit would quantify this and likely tighten the bound substantially.

If the intrinsic fractional width  $r_{\text{int}}$  evolves with energy (e.g. due to changing pair-plasma temperature across epochs), this could partially mimic or mask a foam signal. Marginalizing over a free linear-in- $E$  nuisance term  $r_{\text{int}}(E) = r_0 + r_1(E/E_0)$  leaves the bound essentially unchanged ( $\alpha \gtrsim 0.63$ ), because the  $E^2$  foam scaling has distinct curvature from a linear intrinsic trend across the factor-of-six energy range. Numerically, the best-fit linear slope  $r_1$  is consistent with zero and, although the 95% CL upper limit on  $r_{\text{foam},0}$  broadens from 0.27 to 0.31 when the nuisance is included, the corresponding conservative  $\alpha$  bound shifts by  $\Delta\alpha \approx 3 \times 10^{-3}$  — a direct consequence of the logarithmic compression in the Ng–Perlman relation — confirming that the  $E^2$  and linear- $E$  shapes are effectively orthogonal for the seven-point lever-arm spanning 9–37 MeV.

## B. CMB

For a broadband thermal source such as the CMB, foam does not introduce a spectral-line shift but instead redistributes power: the coherent-core component is suppressed by  $e^{-\sigma_\phi^2(\nu)}$  and the missing flux reappears in a broad pedestal whose shape depends on the short-time behaviour of the correlation function  $C_\alpha(\tau)$ . Because  $\sigma_\phi^2 \propto \nu^2$  (Eq. (9)), the foam deficit is concentrated at high frequencies while the pedestal fills in at low frequencies, producing a spectral signature distinct from standard chemical-potential ( $\mu$ ) and Compton- $y$  distortions.

The Far-InfraRed Absolute Spectrophotometer (FIRAS) aboard the Cosmic Background Explorer (COBE) measured the absolute spectrum of the CMB monopole over 60–600 GHz by direct comparison against an internal cryogenic blackbody calibrator — the absolute-photometry feature that distinguishes it from relative-photometry experiments and makes it well suited to constraining spectral *shape* distortions. The calibrated monopole agrees with a pure blackbody to within  $\sim 50$  ppm (a few  $\times 10^{-5}$  fractional) across the band, placing the still-standard limits  $|\mu| \lesssim 9 \times 10^{-5}$



and  $|y| \lesssim 1.5 \times 10^{-5}$  [18]. These limits and the 43-channel calibrated spectrum form the empirical basis for the foam constraint derived in this subsection, which sets up the broadband signal, derives it from first principles, analyses the FIRAS sensitivity, and contrasts the result with the GRB spectroscopic bound.

### 1. Foam phase variance at CMB frequencies

CMB photons travel from the last-scattering surface at comoving distance  $D_C = 13.87 \text{ Gpc}$  (Planck 2018 flat- $\Lambda$ CDM cosmology;  $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_m = 0.315$ ). For foam path-length variance accumulating uniformly in comoving distance ( $d\sigma_\ell^2/dD_C$  constant, with  $(1+z')^2$  retained inside the integrand of Eq. (11); anchoring (a) of Section II C), the high- $z$  portion of the path dominates the integral and the accumulated phase variance at last scattering is amplified, tightening the FIRAS threshold to  $\alpha \lesssim 0.55$  (Section V B 3;  $\Delta\alpha \approx +0.033$  relative to the simpler form derived below). Because the foam-evolution model in an expanding universe is not specified by Eq. (1), we treat the cosmological enhancement as a multiplicative modification of  $A_\alpha$  that the convention  $A_\alpha = 1$  absorbs.

A more conservative alternative is anchoring (b) of Section II C: foam variance held fixed at the present epoch,  $(1+z')^2 \rightarrow 1$  inside the cosmological integrand, so that Eq. (11) reduces to

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c}\right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (26)$$

where  $\nu$  is the cyclic observer-frame frequency ( $\omega = 2\pi\nu$ ). At  $\nu = 150 \text{ GHz}$  this gives

$$\sigma_\phi^2(150 \text{ GHz}) \approx \begin{cases} 6.8 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \quad (27)$$

The strong  $\nu^2$  dependence concentrates foam effects at high frequency; for  $\alpha = 1/2$  the variance reaches unity at  $\nu \approx 574 \text{ GHz}$ , near the top of the FIRAS band, marking where the linearised template breaks down (Section V B 5). The corresponding FIRAS threshold is  $\alpha \lesssim 0.52$  (Section V B 3), which we adopt as the conservative headline value.

### 2. Foam-modified CMB spectrum

The broadband blackbody admits a direct evaluation: compute  $g_{\text{src}}^{(1)}(\tau)$  as the Fourier transform of  $B_\nu(T_0)$  via Eq. (17), apply the jitter-convolution Eq. (14), and inverse-transform to recover  $S_{\text{obs}}(\nu)$ . This route entangles the model-independent per-mode suppression  $e^{-\sigma_\phi^2(\nu)}$  with the  $C_\alpha$ -dependent redistribution of the suppressed flux. We instead present a derivation that isolates these two contributions: the coherent core, which depends only on the asymptotic phase variance  $\sigma_\phi^2(\nu)$  and is therefore obtained in closed form independent of foam temporal dynamics, and the pedestal kernel  $\hat{R}(\nu - \nu'; \nu')$ , which carries the full dependence on the short- $\tau$  behaviour of  $C_\alpha(\tau)$  and is treated phenomenologically pending a first-principles specification of the foam temporal autocorrelation. A worked example with exponential  $C_\alpha(\tau)$  below makes the pedestal kernel concrete and identifies the physically-realized case used in the Fisher analysis of Section V B 3.

Decomposing the blackbody as an incoherent superposition of narrowband modes  $S_{\text{src}}(\nu') = B_{\nu'}(T_0)$ , each mode is modulated by the *same* achromatic jitter  $\delta t(t)$ , with per-mode foam suppression

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}, \quad \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 \sigma_\ell^2(D), \quad (28)$$

that approaches the constant  $e^{-\sigma_\phi^2(\nu')}$  for all  $\tau \neq 0$  in the  $C_\alpha \rightarrow 0$  limit.

For a single narrowband mode at frequency  $\nu'$ , the source coherence is  $g_{\text{src}}^{(1)}(\tau) = e^{-i2\pi\nu'\tau}$  (the  $\tau_G \rightarrow \infty$  limit of Eq. (15)), and the jitter convolution gives

$$g_{\text{obs}}^{(1)}(\tau) = e^{-i2\pi\nu'\tau} \mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-i2\pi\nu'\tau} e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}.$$

In the dynamical limit,  $\mathcal{F}_{\text{foam}}(\tau; \nu')$  transitions from  $\mathcal{F}_{\text{foam}}(0; \nu') = 1$  to the plateau  $e^{-\sigma_\phi^2(\nu')}$  within the narrow window  $|\tau| \lesssim \tau_{\text{foam}}$ . The Fourier transform of the plateau (constant in  $\tau$  outside that window) is  $e^{-\sigma_\phi^2(\nu')}\delta(\nu - \nu')$  — the coherent core at this mode's frequency — while the Fourier transform of the rise from plateau to unity is supported on  $|\tau| \lesssim \tau_{\text{foam}}$  and is therefore broad in frequency, of width  $\sim 1/\tau_{\text{foam}}$  — the pedestal at this mode. The split applied below to  $\mathcal{F}_{\text{foam}}(\tau; \nu')$  generalises this single-mode decomposition to the full Planck spectrum by linear superposition.

To isolate the coherent-core and pedestal contributions, we split  $\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')} + [\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}]$ : the first term contributes the constant asymptote at all lags (Fourier-transforming to a delta function at  $\nu = \nu'$ ), while the residual is localised at  $|\tau| \lesssim \tau_{\text{foam}}$  (Fourier-transforming to a broad pedestal). Taking the intensity-weighted Fourier transform via Eq. (17) yields

$$S_{\text{obs}}(\nu) = e^{-\sigma_\phi^2(\nu)} B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu', \quad (29)$$

where  $\hat{R}$  denotes the residual's Fourier transform. The coherent-core term  $e^{-\sigma_\phi^2(\nu)} B_\nu(T_0)$  is exact and model-independent (requiring only achromatic jitter and  $C_\alpha \rightarrow 0$ ); the pedestal kernel  $\hat{R}$  depends on the short- $\tau$  behaviour of  $C_\alpha(\tau)$  — the foam coherence timescale  $\tau_{\text{foam}}$  — and is therefore undetermined without specifying  $C_\alpha$ . This is the broadband generalisation of the core-plus-pedestal decomposition stated in Section IV. Because the per-mode phase modulation  $e^{i\omega\delta t(t)}$  is unitary, any physical pedestal kernel  $\hat{R}$  derived from a specific  $C_\alpha(\tau)$  conserves photon number per mode:  $\int [S_{\text{obs}}(\nu) - B_\nu(T_0)] d\nu / (h\nu) = 0$ .

To make the pedestal kernel concrete, we evaluate  $\hat{R}$  for the minimal one-parameter model  $C_\alpha(\tau) = e^{-|\tau|/\tau_{\text{foam}}}$ , the autocorrelation of a Markovian metric fluctuation. The framework does not require this specific choice; any  $C_\alpha(\tau)$  with characteristic decorrelation time  $\tau_{\text{foam}} \ll 1/\nu_{\text{max}}$  (satisfied with enormous margin for Planck-scale foam,  $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44}$  s) has a Fourier transform of width  $\gtrsim 1/\tau_{\text{foam}} \gg \nu_{\text{max}}$ , so the pedestal kernel is constant across the FIRAS band irrespective of its functional form, and the conclusion that the in-band pedestal is negligible follows from the bandwidth ratio alone. Substituting into Eq. (6) gives

$$\sigma_\Delta^2(\tau) = \frac{2\sigma_\ell^2(D)}{c^2} [1 - e^{-|\tau|/\tau_{\text{foam}}}] \quad (30)$$

The Fisher analysis below operates in the linearised regime  $\sigma_\phi^2(\nu') \ll 1$ . Expanding  $\mathcal{F}_{\text{foam}}(\tau; \nu')$  to first order and subtracting the long-lag asymptote  $e^{-\sigma_\phi^2(\nu')}$  leaves the short- $\tau$  residual

$$\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')} \approx \sigma_\phi^2(\nu') e^{-|\tau|/\tau_{\text{foam}}}, \quad (31)$$

whose Fourier transform is a Lorentzian centred on the source frequency,

$$\hat{R}(\nu - \nu'; \nu') = \sigma_\phi^2(\nu') \frac{2\tau_{\text{foam}}}{1 + [2\pi(\nu - \nu')\tau_{\text{foam}}]^2}, \quad (32)$$

of full width at half maximum  $1/(\pi\tau_{\text{foam}})$ . For Planck-scale foam  $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44}$  s the Lorentzian width is  $\sim 6 \times 10^{42}$  Hz, eleven orders of magnitude beyond the FIRAS upper edge  $\nu_{\text{max}} = 6 \times 10^{11}$  Hz; the fraction of the redistributed flux falling within the FIRAS band is  $\sim 10^{-31}$ . The physical pedestal therefore deposits no detectable in-band power for any plausible  $\tau_{\text{foam}}$ . The Fisher analysis of Section V B 3 consequently adopts the no-pedestal limit as the primary template.

To probe robustness against in-band pedestal contamination beyond the physical-kernel result above, we also evaluate the Fisher matrix with a band-filling redistribution kernel deliberately chosen to share the Rayleigh–Jeans  $1/\nu$  shape of the  $\mu$ -distortion template (Section V B 3), thereby maximizing the foam- $\mu$  Fisher degeneracy and yielding the loosest bound any one-parameter smooth pedestal kernel could give:

$$K_\nu^{\text{fast}}(\nu) = \begin{cases} \frac{1}{\nu \ln(\nu_{\text{max}}/\nu_{\text{min}})} & \nu_{\text{min}} \leq \nu \leq \nu_{\text{max}}, \\ 0 & \text{otherwise,} \end{cases} \quad (33)$$

with  $\nu_{\text{min}} = 60$  GHz and  $\nu_{\text{max}} = 600$  GHz set by the FIRAS band edges.  $K_\nu^{\text{fast}}(\nu)$  is a normalized density in observer frequency ( $\int_{\nu_{\text{min}}}^{\nu_{\text{max}}} K_\nu^{\text{fast}} d\nu = 1$ , units  $\text{Hz}^{-1}$ ) that specifies the fraction of the total foam-removed flux reappearing as pedestal intensity per Hz at observer frequency  $\nu$ ; the form above corresponds to uniform energy redistribution in  $\ln \nu$  across the band. Within the FIRAS band,  $K_\nu^{\text{fast}}(\nu)$  replaces the physical pedestal contribution to Eq. (29) by an adversarial factorised substitution

$$\int B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu' \longrightarrow \sigma_\ell^2(D) K_\nu^{\text{fast}}(\nu) P_{\text{foam}}, \quad P_{\text{foam}} \equiv \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} \left( \frac{2\pi\nu'}{c} \right)^2 B_{\nu'}(T_0) d\nu', \quad (34)$$

with  $P_{\text{foam}}$  the total foam-removed flux per unit  $\sigma_\ell^2$ . The substitution forces the pedestal kernel to factorise into a  $\nu$ -only template (the placeholder  $K_\nu^{\text{fast}}$ ) times a  $\nu'$ -only weighting, a structure no physical  $\hat{R}(\nu - \nu'; \nu')$  satisfies (the Lorentzian above carries  $\nu'$ -dependence through  $\sigma_\phi^2(\nu')$ ). The template is energy-conserving but not photon-number-conserving and serves only as a worst-case Fisher probe of kernel-shape sensitivity.

### 3. Fisher analysis: degeneracy with $\mu$ and $y$

We use  $I_\nu \equiv S_{\text{obs}}(\nu)$  to denote the observed CMB intensity including the foam signal. Working in the linearised regime  $\sigma_\phi^2(\nu) \ll 1$ , we treat four spectral parameters  $\theta = (A_{\text{eff}}, T_0, \mu, y)$ , where  $A_{\text{eff}} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha} = \sigma_\ell^2(D_C)$  is the cumulative foam path-length variance at the CMB distance (Eq. (1)), so that  $\sigma_\phi^2(\nu) = (2\pi\nu/c)^2 A_{\text{eff}}$ . Note that  $\alpha$  is not fitted directly; only the lumped amplitude  $A_{\text{eff}}$  enters the Fisher matrix, and bounds on  $\alpha$  are obtained *a posteriori* by inverting the Ng–Perlman relation at fixed  $A_\alpha = 1$ . The foam template is

$$\frac{\partial I_\nu}{\partial A_{\text{eff}}} = - \left( \frac{2\pi\nu}{c} \right)^2 B_\nu(T_0) + K_\nu^{\text{fast}}(\nu) \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} \left( \frac{2\pi\nu'}{c} \right)^2 B_{\nu'}(T_0) d\nu', \quad (35)$$

Equation (35) comprises a high-frequency deficit and a compensating pedestal whose shape is set by the redistribution kernel  $K_\nu^{\text{fast}}(\nu)$  defined in Eq. (33). With  $K_\nu^{\text{fast}} \propto 1/\nu$ , the pedestal integral  $P_{\text{foam}} \equiv \int (2\pi\nu'/c)^2 B_{\nu'}(T_0) d\nu'$  becomes a frequency-independent constant, yielding a  $1/\nu$  pedestal — the same Rayleigh–Jeans shape as the  $\mu$ -distortion template, and hence the origin of the foam– $\mu$  degeneracy.

The chemical-potential and Compton- $y$  distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0) \frac{x e^x}{(e^x - 1)^2}, \quad (36)$$

$$\frac{\partial I_\nu}{\partial y} = B_\nu(T_0) \frac{x e^x}{e^x - 1} \left[ \frac{x(e^x + 1)}{e^x - 1} - 4 \right], \quad (37)$$

with  $x = h\nu/(k_B T_0)$ , encoding respectively early-universe energy injection (Silk damping, decaying particles) and late-time electron heating (reionization, cluster gas). In the Rayleigh–Jeans limit the  $\mu$ -template approaches  $\Delta I/B \rightarrow -\mu/x \propto \nu^{-1}$ , the same frequency dependence as the fiducial pedestal  $K_\nu^{\text{fast}}$ ; the two templates therefore have nearly identical spectral shapes across the FIRAS band, producing a strong Fisher-matrix degeneracy.

A Fisher analysis using the 43 FIRAS channels [18] with diagonal noise approximated as a flat  $\sigma_\nu = 0.01$  MJy/sr per channel (a band-averaged simplification of the published channel-by-channel uncertainties) yields the correlation matrix for parameters  $(A_{\text{eff}}, T_0, \mu, y)$  in that order:

$$\rho_{ij} = \begin{pmatrix} 1.000 & 0.937 & \mathbf{0.976} & 0.010 \\ 0.937 & 1.000 & 0.969 & -0.212 \\ \mathbf{0.976} & 0.969 & 1.000 & -0.127 \\ 0.010 & -0.212 & -0.127 & 1.000 \end{pmatrix}, \quad (38)$$

showing the dominant foam– $\mu$  degeneracy ( $\rho_{A_{\text{eff}}\mu} = 0.976$ , bold) and a negligible foam– $y$  correlation ( $\rho_{A_{\text{eff}}y} = 0.010$ ). Only the  $\sim 5\%$  of foam-direction variance orthogonal to the  $\mu$  template enters the marginal bound; an external constraint on  $\mu$  would directly tighten  $\sigma(A_{\text{eff}})$  by removing this near-degeneracy. Imposing the  $\Lambda$ CDM dissipation prediction  $\mu \approx 2 \times 10^{-8}$  [19] as a prior — some five orders of magnitude tighter than the FIRAS  $\mu$  measurement, hence effectively fixing  $\mu$  — tightens  $\sigma(A_{\text{eff}})$  by  $\sim 3\text{--}5\times$  (less in the no-pedestal limit, where the residual foam– $T_0$  degeneracy dominates), but the logarithmic Ng–Perlman relation converts even this into a threshold gain of only  $\Delta\alpha \lesssim 0.005$ . The marginal Fisher uncertainties are

$$\sigma(A_{\text{eff}}) = 1.7 \times 10^{-12} \text{ m}^2, \quad \sigma(A_{\text{eff}})|_{\mu, y \text{ fixed}} = 2.9 \times 10^{-13} \text{ m}^2, \quad (39)$$

so marginalising over  $\mu$  and  $T_0$  degrades the foam-amplitude uncertainty by a factor  $\sim 6$  (the full  $4 \times 4$  marginal factor exceeds the naive two-parameter estimate  $(1 - \rho_{A\mu}^2)^{-1/2} \approx 4.6$  because of the additional  $T_0$  degeneracy  $\rho_{AT_0} = 0.937$ ), driven almost entirely by the foam– $\mu$  collinearity. Figure 1 plots the SNR versus  $\alpha$ ; setting  $A_{\text{eff}}^{\text{Ng-Perlman}}(\alpha, D_C) = \sigma(A_{\text{eff}})$  and solving for  $\alpha$  yields the threshold  $\alpha \lesssim 0.53$  under the  $K_\nu^{\text{fast}}$  template.

The exponential- $C_\alpha$  worked example of Section V B 2 establishes that the physical pedestal kernel, for any plausible  $\tau_{\text{foam}} \lesssim \ell_P/c$ , displaces essentially all of the redistributed power to frequencies  $\sim 10^{42}$  Hz, far outside the FIRAS band. The no-pedestal limit is therefore the physically-realized case: the foam template degenerates with  $\partial B_\nu/\partial T_0$  at  $\rho \approx 0.9997$  and the sensitivity worsens by  $N \sim 13\times$ , yielding the headline threshold  $\alpha \lesssim 0.52$ . The Fisher matrix above with the adversarial  $K_\nu^{\text{fast}}$  template gives instead  $\alpha \lesssim 0.53$ , a  $\Delta\alpha \approx 0.009$  loosening that quantifies robustness against worst-case in-band pedestal contamination. The near-coincidence reflects the logarithmic Ng–Perlman relation  $\sigma_\ell^2 = D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ : taking the  $\alpha$ -derivative of  $\ln \sigma_\ell^2 = 2(1-\alpha) \ln D_C + 2\alpha \ln \ell_P$  at fixed  $\sigma_\ell$  gives  $d\alpha/d \ln \sigma_\ell = 1/\ln(\ell_P/D_C)$ , so a factor- $N$  worsening in sensitivity shifts  $\alpha$  by only  $\Delta\alpha = \ln N/[2 \ln(D_C/\ell_P)] \approx 0.009$

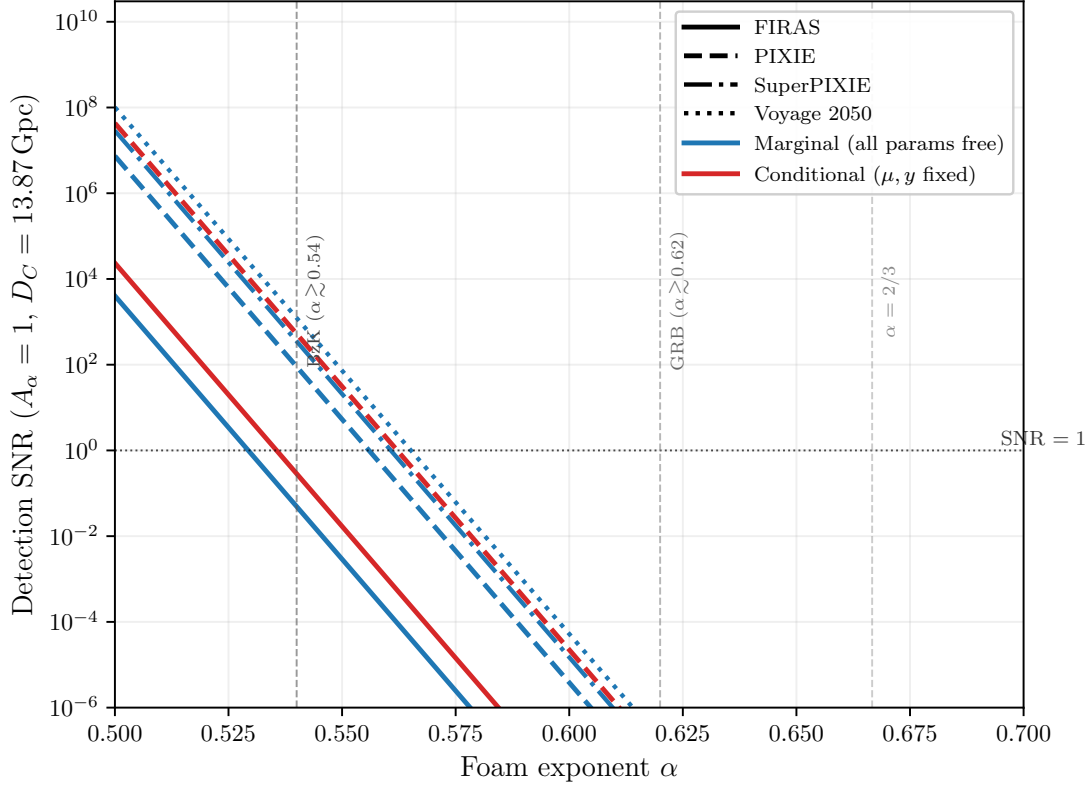


FIG. 1. Detection SNR for the foam amplitude  $A_{\text{eff}}$  as a function of foam exponent  $\alpha$  ( $A_\alpha = 1$ ,  $D_C = 13.87$  Gpc). Blue curves: marginal SNR (all four parameters  $A_{\text{eff}}, T_0, \mu, y$  free). Red curves: conditional SNR ( $\mu, y$  held fixed), which is  $\sim 6\times$  higher. Line styles denote the instrument: solid FIRAS, dashed PIXIE, dash-dotted SuperPIXIE, dotted Voyage 2050, with forecast  $\mu$ -sensitivities scaled relative to FIRAS [19]. The marginal thresholds (SNR = 1) cluster at  $\alpha \approx 0.52$ – $0.56$ , illustrating the logarithmic saturation of the bound with sensitivity. The dotted horizontal line marks SNR = 1; vertical dashed lines mark the spectroscopic lower bound from GRB 221009A ( $\alpha \gtrsim 0.63$ ) and the holographic prediction  $\alpha = 2/3$ .

for  $N \approx 13$ . Either calculation requires specifying  $C_\alpha(\tau)$  from first principles for a definitive in-band pedestal shape, but the worked example shows that any exponential-class  $C_\alpha$  with Planck-scale  $\tau_{\text{foam}}$  lies in the no-pedestal regime.

The proposed PIXIE satellite [20] — a fully cryogenic ( $T_{\text{sys}} \approx 2.7$  K) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching  $|\mu| \lesssim 10^{-8}$  and  $|y| \lesssim 10^{-9}$  and entering the regime of guaranteed  $\Lambda$ CDM spectral signals from Silk damping and reionization. Propagating this gain through the Fisher-matrix scaling derived above, the marginal foam detection threshold shifts from  $\alpha \lesssim 0.52$  (FIRAS) to approximately  $\alpha \lesssim 0.55$  (PIXIE), narrowing the gap with the spectroscopic lower bound  $\alpha \gtrsim 0.63$  from GRB 221009A (Fig. 1). Both FIRAS and PIXIE thresholds are quoted for the placeholder  $K_\nu^{\text{fast}} \propto 1/\nu$  pedestal kernel and the observer-frame  $\sigma_\phi^2$  of Eq. (9); a physically derived  $C_\alpha(\tau)$  would generally relax the foam- $\mu$  degeneracy and tighten these bounds, while the cosmological-redshift weighting of Eq. (11) could shift each by up to  $\Delta\alpha \sim 0.03$  in the same direction. The dominant foam- $\mu$  degeneracy ( $\rho \approx 0.98$ ) persists on the linearized template, but PIXIE’s extension to 6 THz accesses the nonlinear regime  $\sigma_\phi^2 \gg 1$  for  $\alpha \lesssim 0.55$  under the cosmological-redshift weighting of Eq. (11) (observer-frame Eq. (9) shifts this to  $\alpha \lesssim 0.52$ ), where the foam template diverges markedly from the  $\mu$ -distortion shape and the degeneracy may be substantially reduced. The  $\alpha \lesssim 0.55$  projection is therefore conservative: if the nonlinear template effectively breaks the foam- $\mu$  degeneracy at PIXIE sensitivity, the true threshold could be significantly more stringent.

#### 4. Bias in recovered $T_0$ from unmodeled foam

The Fisher matrix also quantifies how much the fitted CMB temperature would shift if foam were present but omitted from the model. The magnitude and sign of the resulting  $T_0$  bias both depend on the assumed pedestal kernel  $K_\nu^{\text{fast}}$ , which is a placeholder rather than framework-derived (Section V B 2); a physical kernel from a specified

$C_\alpha(\tau)$  could alter either. Suppose the data contain foam at amplitude  $A_{\text{eff}}^{\text{true}}$  but the analyst fits only  $\theta_{\text{fit}} = (T_0, \mu, y)$ . In the linearised Fisher regime the parameter shift is

$$\hat{\theta}_{\text{fit}} - \theta_{\text{fit}}^{\text{true}} = -(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}} A_{\text{eff}}^{\text{true}}, \quad (40)$$

where  $F_{\text{fit,fit}}$  is the  $3 \times 3$  Fisher block for  $(T_0, \mu, y)$  and  $F_{\text{fit,omit}}$  couples the fitted parameters to the omitted foam amplitude. Both blocks come from the same matrix as Eq. (38), so no new computation is required.

The  $T_0$  component gives

$$\Delta T_0 = b_{T_0} A_{\text{eff}}^{\text{true}}, \quad b_{T_0} = -[(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}}]_{T_0}. \quad (41)$$

Numerically,  $b_{T_0} = +1.21 \times 10^7 \text{ K m}^{-2}$ : the bias is *positive*, reflecting the placeholder pedestal's Rayleigh–Jeans excess being absorbed as an apparent temperature increase ( $\partial B_\nu / \partial T > 0$ ), while the high-frequency deficit is absorbed into a positive  $\Delta\mu$  ( $b_\mu \approx 1.05 \times 10^8 \text{ m}^{-2}$ , consistent with  $\rho_{A_{\text{eff}}\mu} = 0.976$ ). Requiring  $|\Delta T_0| < \sigma(T_0) = 5.7 \times 10^{-4} \text{ K}$  [21] gives

$$A_{\text{eff}}^{\text{max, bias}} = \frac{\sigma(T_0)}{|b_{T_0}|} = 4.7 \times 10^{-11} \text{ m}^2, \quad (42)$$

corresponding (with  $A_\alpha = 1$ ) to a crossover exponent  $\alpha_{\text{bias}} = 0.518$ . Models with  $\alpha < 0.518$  would have biased the Fixsen  $T_0$  by more than  $1\sigma$  had such a signal been present and unmodeled; models with  $\alpha > 0.518$  would not.

The crossover  $\alpha_{\text{bias}} \approx 0.518$  essentially coincides with the marginal detection threshold  $\alpha \lesssim 0.52$ : with the correctly normalized noise the foam amplitude that FIRAS could just detect is also large enough to shift  $T_0$  by of order  $1\sigma(T_0)$ . The two boundaries probe complementary regimes from the GRB lower exclusion bound  $\alpha \gtrsim 0.63$  — FIRAS sets an upper detectability limit while the GRB analysis sets a lower exclusion bound. Mechanistically, the foam pedestal's Rayleigh–Jeans-like low-frequency excess overlaps the  $\mu$ -distortion template much more than the  $\partial B_\nu / \partial T_0$  template, so in the four-parameter fit the foam signal is absorbed primarily into  $\mu$ , leaving  $T_0$  only weakly shifted; this is why the  $T_0$ -bias crossover ( $\alpha \approx 0.518$ ) lies marginally below, rather than far below, the detection threshold. Figure 2 plots  $|\Delta T_0|/\sigma(T_0)$  as a function of  $\alpha$ ; at the detection threshold  $\alpha \approx 0.52$  (marginal SNR  $\approx 1$ ) the ratio is  $\approx 0.5$ , falling below  $10^{-2}$  only for  $\alpha \gtrsim 0.54$ . The published Fixsen  $T_0 = 2.72548 \text{ K}$  is therefore unbiased at the  $\ll 10^{-4}\sigma$  level by any foam model consistent with the GRB bound  $\alpha \gtrsim 0.63$ , although a hypothetical strong-foam model at the very edge of FIRAS detectability ( $\alpha \approx 0.52$ ) would shift it by up to  $\sim 0.5\sigma(T_0)$ .

### 5. Caveats: nonlinear regime and parameter degeneracy

Two caveats qualify the linearised analysis.

*Nonlinear regime.* The linearisation  $\Delta I_\nu \approx \sigma_\phi^2(\nu) \times [\text{template}]$  is valid only for  $\sigma_\phi^2 \ll 1$ . For  $\alpha = 1/2$ ,  $\sigma_\phi^2 = 1$  is reached at  $\nu \approx 574 \text{ GHz}$ , within the FIRAS band. Replacing the linearised deficit  $-\sigma_\phi^2(\nu)B_\nu$  by the saturated form  $-(1 - e^{-\sigma_\phi^2(\nu)})B_\nu$  reduces the *integrated* in-band foam amplitude (summed over FIRAS channels) by  $\approx 13\%$  at  $\alpha = 0.50$  and by  $< 1\%$  at  $\alpha \gtrsim 0.52$ . At the headline threshold  $\alpha \lesssim 0.52$  the saturation correction is thus only  $\approx 1\%$  ( $\Delta\alpha \sim 3 \times 10^{-5}$ ), so the linearised bound is unaffected at the  $\Delta\alpha \ll 10^{-3}$  level; the nonlinear template becomes important only for stronger foam ( $\alpha \lesssim 0.50$ ), where it may also help break the foam– $\mu$  degeneracy.

*Parameter degeneracy within the foam template.* The exponent  $\alpha$  and the amplitude  $A_\alpha$  enter only through the product  $A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha} = A_{\text{eff}}$ . FIRAS constrains this product directly but not the exponent separately; converting a bound on  $A_{\text{eff}}$  to a bound on  $\alpha$  requires fixing  $A_\alpha$  by convention (here  $A_\alpha = 1$ , following Ng & Perlman [4]). Any quoted bound on  $\alpha$  carries this implicit normalisation.

### 6. Complementarity with GRB 221009A

The FIRAS sensitivity range  $\alpha \lesssim 0.52$ – $0.55$  does not overlap the GRB spectroscopic lower bound  $\alpha \gtrsim 0.63$  of Section V A: FIRAS probes strictly lower exponents. Any foam model consistent with the GRB bound has  $\sigma_\phi^2 \ll 1$  at CMB frequencies, so the  $T_0$  bias is entirely negligible (Section V B 4 and Fig. 2).

FIRAS provides an independent broadband test of foam — absolute calibrated photometry, no source-profile assumption, and a different systematic-error budget from spectroscopy — and constrains the parameter space below the GRB bound where more exotic foam models would reside. A definitive FIRAS foam analysis would require specifying  $C_\alpha(\tau)$  to determine the physical pedestal shape and replacing the linearised template with the full nonlinear

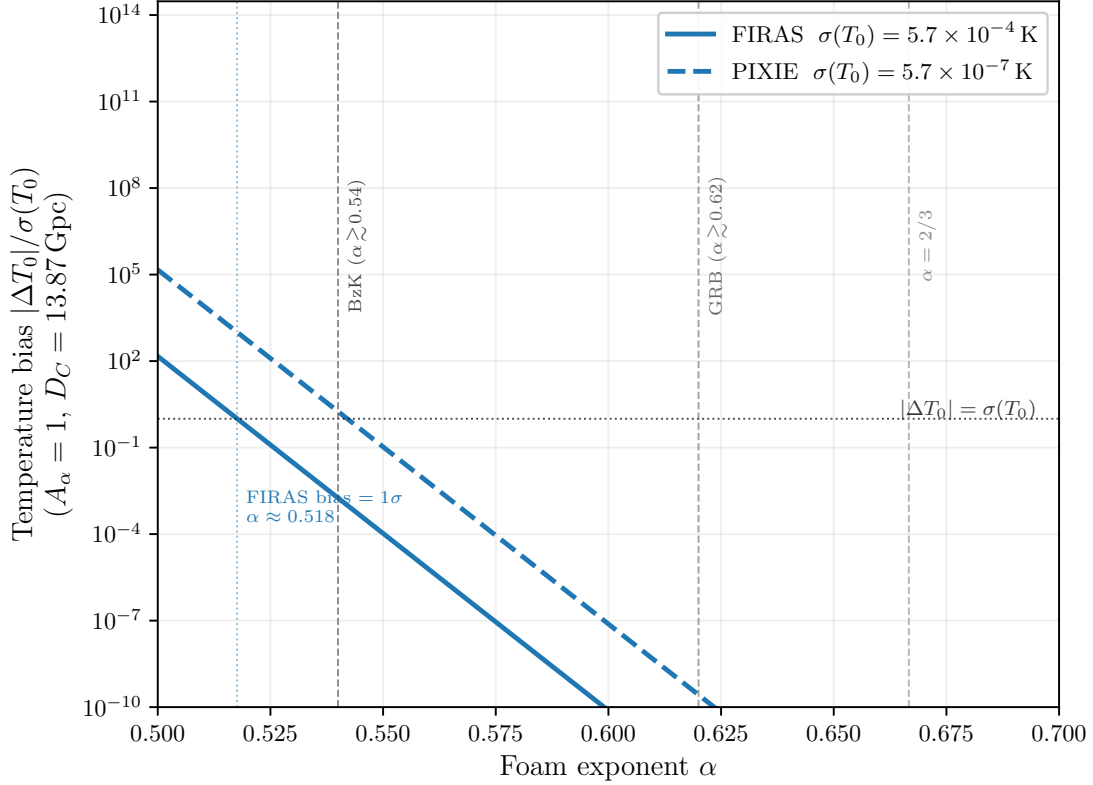


FIG. 2. Foam-induced bias in the recovered CMB temperature,  $|\Delta T_0|/\sigma(T_0)$ , as a function of foam exponent  $\alpha$  when foam at amplitude  $A_{\text{eff}}$  ( $A_\alpha = 1$ ,  $D_C = 13.87 \text{ Gpc}$ ) is present but omitted from the spectral fit. Solid: FIRAS with  $\sigma(T_0) = 5.7 \times 10^{-4} \text{ K}$  [21]; dashed: PIXIE with  $\sigma(T_0) \approx 5.7 \times 10^{-7} \text{ K}$  (noise floor  $\sim 10^3 \times$  lower). The dotted horizontal line marks  $|\Delta T_0| = \sigma(T_0)$ ; the FIRAS crossover occurs at  $\alpha \approx 0.518$ . Vertical dashed lines mark the spectroscopic lower bounds and the holographic prediction  $\alpha = 2/3$  (as in Fig. 1).

expression at high frequencies where  $\sigma_\phi^2 \gtrsim 1$ . The next generation of absolute spectrophotometers would deepen this reach only marginally. Extending the Fisher analysis above to the 10 GHz–3 THz band of the SuperPIXIE and Voyage 2050 concepts [19] — with their forecast distortion sensitivities  $\sigma(\mu) \approx 7.7 \times 10^{-9}$  and  $2.2 \times 10^{-9}$ , respectively — shifts the marginal foam threshold only to  $\alpha \lesssim 0.55$  (SuperPIXIE) and  $\alpha \lesssim 0.56$  (Voyage 2050), barely beyond the PIXIE projection. The reason is structural rather than instrumental: in the physically-realized no-pedestal limit the foam deficit  $-\sigma_\phi^2(\nu)B_\nu$  is degenerate with a temperature shift at  $\rho_{AT_0} = 0.9997$ , and this degeneracy is *band-robust* — broadening the coverage from the FIRAS band to 10 GHz–6 THz leaves it unchanged to four decimal places and even slightly worsens the foam-to- $\mu$  sensitivity ratio  $\sigma(A_{\text{eff}})/\sigma(\mu)$ . The wider band does break the foam- $\mu$  degeneracy probed by the adversarial  $K_\nu^{\text{fast}}$  pedestal (from  $\rho_{A\mu} = 0.98$  at FIRAS to 0.66 over 10 GHz–3 THz), so future missions are markedly more robust against worst-case in-band pedestal contamination; but this does not move the physical threshold. Together with the logarithmic Ng–Perlman scaling  $\Delta\alpha = \ln N/[2 \ln(D_C/\ell_P)]$ , which converts even a  $10^4$ -fold sensitivity gain over FIRAS into  $\Delta\alpha \lesssim 0.04$ , this places a firm ceiling: spectral-distortion photometry probes the foam parameter space at  $\alpha \lesssim 0.56$  and cannot, by sensitivity or bandwidth alone, reach the GRB spectroscopic bound  $\alpha \gtrsim 0.63$  or the holographic value  $\alpha = 2/3$ . The two channels are therefore permanently complementary rather than overlapping.

## VI. CONCLUSIONS

All single-source spectroscopic bounds derived in this paper assume the dynamical-foam limit  $\tau_{\text{foam}} \ll \tau_c$  (Section III B), in which the foam path-length jitter decorrelates many times within each photon coherence time. If foam fluctuations are instead slow ( $\tau_{\text{foam}} \gg \tau_c$ ), the accumulated phase is essentially frozen during each coherence time and the foam acts as a deterministic, frequency-independent phase screen — no spectral broadening or peak reduction occurs within a single observation, and the shape-distortion bounds do not apply. The CMB constraint (Section V B)

is similarly derived in the dynamical limit (Section VB2 works in  $C_\alpha \rightarrow 0$ ): in the opposite quasi-static limit the per-mode foam suppression  $\mathcal{F}_{\text{foam}}(\tau) \rightarrow 1$  (Eq. (10)) and the CMB monopole is unmodified, so the shape-distortion bounds do not apply in either channel.

The dynamical bound  $\alpha \gtrsim 0.63$  (Eq. (25)) does not exclude the holographic value  $\alpha = 2/3 \approx 0.667$ . A principal instrumental limitation of the present analysis is the MeV sensitivity gap: the spectral region between the upper edge of the Fermi-GBM BGO detector response ( $\lesssim 40$  MeV, where the effective area falls steeply) and the regime in which Fermi-LAT achieves robust detection efficiency ( $\gtrsim 100$  MeV). The frequency  $\nu_* \simeq 37$  MeV at which the foam-induced spectral break is predicted to be most constraining falls squarely within this gap, so neither instrument provides fully resolved coverage across the critical transition. Closing the MeV gap is a recognized community priority, addressed by next-generation satellite concepts that combine Compton-scattering and pair-conversion detection techniques to achieve continuous sensitivity from  $\sim 100$  keV to beyond 1 GeV. The All-sky Medium Energy Gamma-ray Observatory (AMEGO [22]) and its MIDEX-scale variant AMEGO-X [23] are designed to cover 100 keV to 1 GeV with order-of-magnitude improvements over both COMPTEL and Fermi-LAT in the MeV decade; the ESA-proposed M-class concept e-ASTROGAM [24] targets a comparable bandpass. The near-term NASA SMEX mission COSI (scheduled launch  $\sim 2027$ ) will extend broadband sensitivity to 0.2–5 MeV, providing a valuable pathfinder, though its upper energy bound falls well short of the 37 MeV regime and the gap therefore remains open. Realization of any of the broader-bandpass concepts would, for the first time, permit fully resolved spectral measurements of GRB prompt emission across the 37 MeV transition without the instrumental discontinuity that limits the present work; reducing the per-epoch line-width uncertainties to  $\delta r \lesssim 0.01$  would tighten the bound to  $\alpha \gtrsim 0.64$ , removing a major systematic and enabling substantially tighter tests of spacetime-foam models.

The multi-epoch  $r_{\text{obs}}$  dataset constrains any propagation effect that broadens an emission line in proportion to a power of  $E$ , not only the Ng–Perlman spacetime-foam template considered here. The most natural extension is to stochastic Lorentz-invariance violation [13], in which the photon group velocity carries a Planck-suppressed random coefficient  $v(E) = c[1 - \xi(E/E_{\text{QG},n})^n]$ , with  $\xi$  of order unity drawn independently for each Planck-scale traversal. Photons in the same line then propagate at slightly different speeds, broadening the observed width as  $\sigma_{\text{LIV}}(E)/E \propto E^n$  on top of any intrinsic contribution. Unlike the deterministic LIV signal probed by photon time-of-flight studies — where GRB 221009A LHAASO TeV photons already constrain  $E_{\text{QG},1} \gtrsim \text{a few} \times E_{\text{Pl}}$  [25] — stochastic LIV produces a *width* signal that survives light-curve cross-correlation, and a multi-epoch  $r_{\text{obs}}$  measurement is therefore one of the few clean probes available. Fitting Eq. (24) recovers the foam constraint as the  $n = 1$  case; refitting the same seven-epoch dataset under  $r_{\text{LIV}} \propto (E/E_0)^n$  would yield bounds on  $\langle \xi^2 \rangle^{1/2}/E_{\text{QG},n}^n$  at each order, with the  $n = 2$  (quadratic) case particularly valuable because the deterministic time-of-flight tests are weakest there at  $z = 0.151$ .

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